

Structure–Function Relationships in Factor IX and Factor IXa

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Factor IX (FIX) consists of an N-terminal γ -carboxyglutamic acid (Gla) domain followed by two epidermal growth factor (EGF)-like domains, and the C-terminal serine protease domain. During physiologic coagulation, one of the activators of FIX is the FVIIa/tissue factor (TF) complex. In this reaction, the Gla and EGF1 domains of FIX are thought to interact with TF. The FIXa that is generated then combines with FVIIIa on the platelet surface to activate FX in the coagulation cascade. In this assembly, the protease domain and possibly the EGF2 domain of FIXa are thought to provide the primary specificity in binding to FVIIIa. Disruption of the interaction of FIX/FX with TF and of the FIXa:FVIIIa interface may provide a pharmacologic target as an alternative strategy for the development of antithrombotic agents. (Trends Cardiovasc Med 2003;13:39–45). © 2003, Elsevier Science Inc.

Factor IX (FIX) plays an important role in blood coagulation because absence or deficiency of its activity results in an X-linked bleeding diathesis, hemophilia B. The major proteins involved in the initiation of coagulation include tissue factor (TF), FVIIa, FIX, and FX (Hockin et al. 2002). As illustrated in Figure 1, physiologic coagulation begins by exposure of TF in the extravascular space and its association with small amounts of FVIIa present in blood (Morrissey et al. 1993). The TF/FVIIa complex formed activates both FIX and FX. In a feedback loop, FIXa, FXa, and TF/FVIIa generate additional amounts of TF/FVIIa from TF/

FVII, which serves to augment the initiation phase of coagulation. The FXa formed also generates small amounts of thrombin, which serves to activate FV and FVIII (Hoffman et al. 1995). The FIXa and FVIIIa thus generated assemble on the platelet surface to activate substantial amounts of FX to FXa (Butenas et al. 2001). The FXa then combines with FVa to enhance thrombin generation and fibrin formation (Davie et al. 1991, Hoffman et al. 1995). Thrombin may also activate FXI to FXIa on the platelet surface by a feedback mechanism to allow further activation of FIX to FIXa for sustained and consolidated coagulation (Baird and Walsh 2002). In the current scheme of coagulation, activation of FIX by TF/FVIIa and of FX by FIXa is important because initial FXa generated by the **TF/FVIIa complex combines with TF pathway inhibitor (TFPI)** and the resultant FXa/TFPI shuts down TF/FVIIa activity (Bajaj et al. 2001a, Girard et al. 1989). Thus, the FXa generated by the FIXa/FVIIIa complex represents a key step in coagulation, and disruption of this step may represent a potential target for development of new antithrombotics.

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• FIX Structure

FIX is a vitamin K-dependent protein that is synthesized by hepatocytes as a precursor of a serine protease, FIXa. The gene for FIX consists of eight exons and seven introns, is approximately 34 kb long, and is located on the long arm of the X chromosome at Xq27.1 (reviewed in Thompson 2001). It is synthesized as a precursor protein of 461 amino acids containing a 28-residue signal prepeptide and an 18-residue leader propeptide. During biosynthesis, the gene for FIX undergoes several posttranslational modifications. These include γ -carboxylation of the first 12 Glu residues; glycosylation at Asn-157, Asn-159, Ser-53, Ser-61, Thr-159, Thr-169, and Thr-172; partial hydroxylation of Asp-64; sulfation of Tyr-155; phosphorylation of Ser-158; and removal of the signal and the propeptide sequences (Thompson 2001). The resulting mature protein is a single chain of 415 residues (Mr 57,000) that contains approximately 17% carbohydrate by weight (Di Scipio et al. 1978).

Structurally, FIX contains an N-terminal Gla domain (residues 1–40), a short hydrophobic stack (residues 41–46), two epidermal growth factor (EGF)-like domains (EGF1: residues 47–83, and EGF2: residues 88–127, which are connected by linker residues 84–87), an activation peptide (residues 146–180), and a C-terminal protease domain (residues 181–415) (Brandstetter et al. 1995, Hopfner et al. 1999). The Gla domain possesses several intermediate- and low-affinity Ca^{2+} binding sites (Amphlett et al. 1981, Bajaj 1982, Li et al. 1997), and the EGF1 and protease domain each possess a single high-affinity Ca^{2+} binding site (Bajaj et al. 1992, Rao et al. 1995). Ca^{2+} binding to each domain imparts structural properties to FIX that confer biologic function (Freedman et al. 1995, Lenting et al. 1996, Mathur et al. 1997). The domain structure of FIX and the Ca^{2+} sites in each domain are illustrated in Figure 2.

• Activation of FIX

As stated earlier, FIX can be activated to FIXa by either the TF/FVIIa/ Ca^{2+} complex or FXIa/ Ca^{2+} (Di Scipio et al. 1978, Østerud and Rapaport 1977). This activation involves proteolytic cleavages in FIX at the Arg¹⁴⁵–Ala¹⁴⁶ and Arg¹⁸⁰–Val¹⁸¹

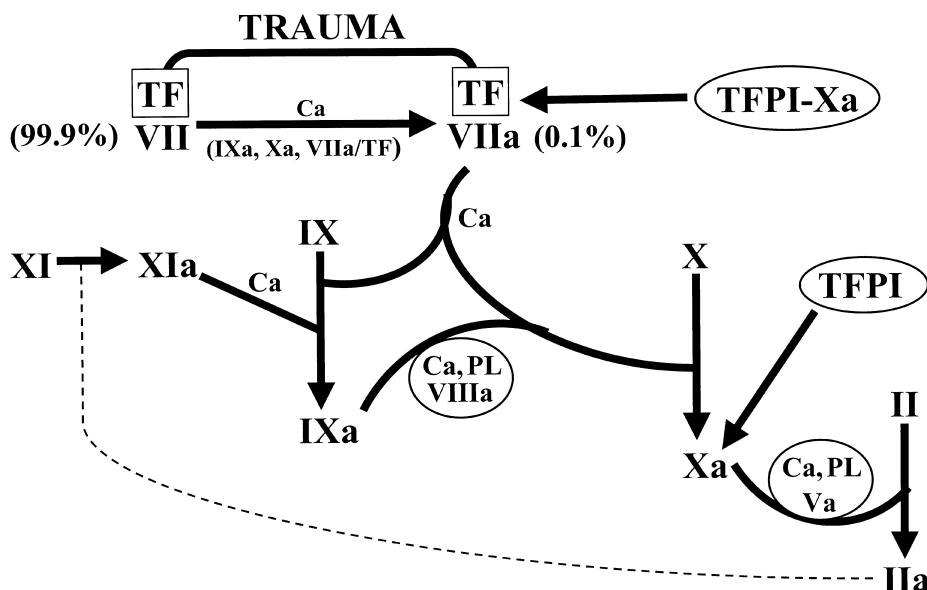


Figure 1. The current scheme of blood coagulation. Coagulation is initiated by exposure of blood to TF and formation of a Ca^{2+} -dependent complex between TF and FVII/FVIIa. The TF/FVIIa complex directly as well as indirectly (through the FIXaFVIIIa complex) activates FX. The initial FXa formed via direct activation by the TF/FVIIa complex generates small amounts of thrombin, which activates the two key cofactors, FVIII and FV, for their assembly with FIXa and FXa on the platelet surface, respectively. The direct activation of FX is rapidly shut down by TFPI (and possibly by antithrombin/heparin), and the indirect generation of FXa via FIXa is necessary for effective coagulation to proceed. FIXa generated via FXIa further aids in consolidation of the coagulation cascade. Activation of FXI to FXIa is shown by the dotted line. For clarity, thrombin activation of FVIII and FV is not shown. An approximate percentage ratio of FVII to FVIIa present in circulating blood is indicated. Ca, calcium; F, factor; PL, platelet phospholipid membrane; II, prothrombin; IIa, thrombin; TF, tissue factor; TFPI, tissue factor pathway inhibitor.

bonds with a concomitant release of a 35-residue activation peptide (Di Scipio et al. 1978). During this process, the Arg¹⁴⁵-Ala¹⁴⁶ peptide bond in FIX appears to be cleaved first, followed by cleavage of the Arg¹⁸⁰-Val¹⁸¹ bond (Bajaj et al. 1983, Di Scipio et al. 1978, Lawson and Mann 1991). The FIXa thus formed contains a light chain (residues 1–145) and a heavy chain (residues 181–415) held together by a single disulfide bond. The light chain consists of the Gla, EGF1, and EGF2 domains, whereas the heavy chain contains the serine protease domain that features the catalytic triad of residues Ser-c195 (365), His-c57 (221), and Asp-c102 (269). For comparison, the chymotrypsin amino acid numbering system is used for the protease domain of FIXa. Thus, the numbers with the prefix “c” (e.g., c195) refer to the chymotrypsin equivalents for the protease domain of FIXa. The corresponding residues in parentheses represent FIX numbering.

Efforts have been directed toward understanding the regions in TF/FVIIa

that interact with FIX (or FX) during its activation. A series of kinetic experiments related to the activation of FX by the TF/FVIIa complex and its inhibition pattern by *p*-aminobenzamidine revealed that the exosites remote from the active site of TF/FVIIa determine its affinity for FX and, by inference, for FIX (Baugh et al. 2000). In support of this concept, initial studies performed revealed that the **EGF1 domain is required for FIX activation by the TF/FVIIa complex but not by FXIa (Zhong et al. 1994)**. Further studies by Wu et al. (2000) revealed that mutation of Gly-48 in the EGF1 domain of FIX substantially impaired its activation by TF/VIIa. Subsequently, Zhong et al. (2002) provided evidence that the **EGF1 domain of FIX (or FX) interacts with TF in the TF/FVIIa complex**.

In additional studies, Kirchhofer et al. (2000) found that the Gla domains of FIX and FX also participate in binding to the TF/FVIIa complex. For these studies, they mutated several surface-exposed residues in the C-terminal domain of TF. In their modeled activation complex, the

Gla domain of FIX (or FX) is positioned to interact with the C-terminal domain of TF in the TF/FVIIa complex (Kirchhofer et al. 2000). Thus, activation of FIX (or FX) was impaired when the TF/FVIIa complex that was made with the mutated TF was used as an activator. Further, using docking and mutagenesis experiments, Ruf et al. (1999) showed that the Gla domain of FVIIa was important in recognizing the Gla domain of FX. Later studies by Kirchhofer et al. (2001) revealed that the same region of TF that interacts with the FX Gla domain also interacts with FXa during the reciprocal activation of TF/FVII to TF/FVIIa. Thus, in addition to the primary scissile bond recognition sites (Arg-145 and Arg-180) where proteolytic events occur, at least two exosite regions in FIX—namely, the Gla and EGF1 domains—are involved in the TF/FVIIa/FIX ternary complex. Extensive modeling studies by Chen et al. (2002) also support the biochemical data that the Gla and EGF1 domains of FIX are involved in binding to TF in the FIX/TF/FVIIa ternary complex. This is schematically depicted in Figure 2.

• FIXa:FVIIIa Interaction

Several laboratories are actively pursuing the domains in FIXa that interact with FVIIIa. Briefly, FVIIIa is a heterotrimer composed of three noncovalently associated protein chains, namely, A1, A2, and A3-C1-C2 (Fay et al. 1991a, Lollar and Parker 1989). A3-C1-C2 is also referred to as the light chain and is associated with the A1 subunit in a divalent metal ion-dependent linkage, whereas the A2 subunit is weakly associated with the A1 and A3-C1-C2 dimer (Fay et al. 1991b, Lollar and Parker 1990). Based on experimental data from several laboratories, two apparently very similar models (Bajaj et al. 2001b, Stoilova-McPhie et al. 2002) of the FIXa:FVIIIa interaction have been proposed. It is becoming increasingly clear that the protease domain and possibly the EGF2 domain of FIXa bind to FVIIIa by involving the A2 and A3 subunits. In this context, the EGF1 domain may play a structural role in properly positioning the EGF2 and protease domains for binding to FVIIIa. The role of the Gla domain in this context is unclear. The data pertaining to the role of each domain in its interaction with FVIIIa is outlined below.

The first indication that the protease domain of FIXa may be involved in binding to FVIIIa came from the studies of Astermark et al. (1994) and Bajaj et al. (1985). At present, strong evidence exists that residues of the c126-helix (293-helix) as well as those of the c162-helix (330-helix), and residue Asn-c178 (346) of FIXa bind to FVIIIa (Kolkman et al. 1999, Lefkowitz et al. 2001, Mathur and Bajaj 1999). Furthermore, work from the laboratories of Bajaj and Fay (Bajaj et al. 2001b) provides supportive data indicating that the c162-helix in FIXa binds to the 558–565 region of the A2 subunit of FVIIIa. This is schematically depicted in Figure 3. The FIXa:FVIIIa interface shown in this figure is based on the molecular models built previously by Bajaj et al. (2001b) and Stoilova-McPhie et al. (2002), with the inclusion of additional residues from the EGF2 domain (see below).

Mutation of Glu-c80 (245) to Val, a hemophilia B mutation, has also been shown to result in diminished binding of FIXa to FVIIIa (Mathur et al. 1997). However, mutations in the adjacent residues in the Ca²⁺ binding loop have little or no effect on FVIIIa binding (Hamaguchi et al. 1994). The carboxylate of the side chain of Glu-c80 is expected to coordinate to Ca²⁺, and mutation of this residue to Val should abolish the Ca²⁺ binding site in the protease domain (Figure 4). Because binding of FVIIIa is only impaired when Glu-c80 in FIXa is changed to Val and not when non-coordinating residues in this loop are mutated, it is inferred that the protease domain Ca²⁺ binding loop is not directly involved in binding of FIXa to FVIIIa. Instead, it imparts a structural transition that favors FVIIIa binding (Bajaj 1999).

Proteolysis in the autolysis loop—residues c141–c155 (310–323)—of FIXa also results in diminished binding to FVIIIa (Mathur et al. 1997, Schmidt and Bajaj 2001). The mutation of autolysis loop residue Lys-c148 (316) to Ala or Glu (Kolkman and Mertens 2000), as well as proteolysis at Lys-c148/Gly-c149 (Schmidt and Bajaj 2001), results in a substantial reduction in the rate of FX activation both in the presence and absence of FVIIIa. This indicates the importance of the autolysis loop in substrate binding. How does proteolysis at Lys-c148/Gly-c149 (316/317) (Schmidt and Bajaj 2001)

and Arg-c150/Ser-c151 (318/319) (Mathur et al. 1997) adversely affect FVIIIa binding? An explanation that we would like to put forth is that each FVIIIa binding helix (c126-helix and c162-helix) is connected to the autolysis loop by a single β strand, and cleavage of the autolysis loop could allow both of these helices to shift slightly and be misaligned in their interactions with FVIIIa. The spatial relationship between the FVIIIa binding helices, autolysis loop, and the Ca²⁺ loop is depicted in Figure 4.

EGF2 Domain

Recently, several reports have been published pertaining to the role of the EGF2 domain of FIXa in binding to FVIIIa as well as to the platelet surface. Wilkinson et al. (2002b) replaced each of the three EGF2 loops (loop 1: residues 88–99, loop 2: residues 95–109, and loop 3: residues 111–124) with the comparable loops from FVIIa and replaced the entire EGF2 domain with that from FVIIa. They found no apparent difference in the binding of these FIXa chimeras to FVIIIa; however, the apparent K_d for the FX activating complex assembled on the platelet phospholipid membrane (PL) surface was increased for the loop 1, loop 2, and entire EGF2 replacement chimeras, but not for the loop 3 chimera (Wilkinson et al. 2002b). From these data, the authors concluded that residues 88–109 contained in loops 1 and 2 of the EGF2 domain of FIXa are important for assembly of the FX activating complex on the PL surface. In further studies, they investigated the binding of these EGF2 chimeras to the activated platelets and found that residues 88–109 of the EGF2 domain in FIXa mediate binding to platelets and assembly of the FX activating complex (Wilkinson et al. 2002a). Thus, at least from these two studies, one conclusion to be drawn is that the EGF2 domain apparently plays a minor role in binding to FVIIIa.

Celie et al. (2002) also investigated the role of the EGF2 domain of FIXa in binding to FVIIIa. They studied the role of the EGF1/EGF2 linker region Leu⁸⁴–Thr⁸⁷ as well as the Asn⁸⁹–Lys⁹¹ loop region of FIXa in binding to FVIIIa. They substituted these segments with corresponding regions from FX or FVII and found that replacement of the Asn⁸⁹–Lys⁹¹ region had only a minor reduction

in the ability of FIXa to activate FX in the presence of FVIIIa. In contrast, insertions or substitutions of the Leu⁸⁴–Thr⁸⁷ linker region in FIXa greatly affected FX activation in the presence of FVIIIa or its A2 subunit. However, binding of the A2 subunit was only marginally affected. Additionally, in a preliminary report using cross-linking experiments, Olsen et al. (2001) revealed that residues 68–94 of FIXa may bind to the 1804–1818 region of the A3 subunit of FVIIIa. Thus, the FIXa Leu⁸⁴–Thr⁸⁷ linker region may be important in binding to the A3 subunit of FVIIIa. Moreover, because the EGF2 domain makes extensive contact with the protease domain (Hopfner et al. 1999), insertions or substitutions of the Leu⁸⁴–Thr⁸⁷ linker region could affect the conformation of the protease domain and enhancement of its A2 subunit-mediated enzymatic activity without greatly affecting its association with the A2 subunit. Thus, as proposed by Celie et al. (2002), the interaction of the FIXa protease domain with FVIIIa may be allosterically controlled by a mechanism or mechanisms that involve the Leu⁸⁴–Thr⁸⁷ linker region, as well as possible direct contacts between the EGF2 domain and FVIIIa.

The effect of Ala point mutations in the EGF2 domain of FIXa has also been examined (Chang et al. 2002). These authors proposed that residues 89–94 are involved in binding to FVIIIa and that residues 102–108 are involved in binding to FX. The role of the residues 102–108 proposed by Chang et al. (2002) is consistent with the conclusions of Wilkinson et al. (2002a and 2002b), who also proposed that these residues are important in the assembly of the FX activating complex. However, the proposed functional significance of residues 89–94 by Chang et al. (2002) is in contrast to the results of Wilkinson et al. (2002b) and Celie et al. (2002). Chang et al. (2002) presented data indicating complete abolishment of FVIIIa binding for N89A and N92A. However, in the study conducted by Celie et al. (2002) and Wilkinson et al. (2002b) in which N89 was changed to Ser or Val and N92 was unchanged, FVIIIa binding was only marginally affected. The reason(s) for the discrepancy between the observations of Chang et al. (2002) and those of Celie et al. (2002) and Wilkinson et al. (2002b) are not clear. Further studies

are needed to resolve the differences and unequivocally establish the role of N89 in binding to FVIIIa. As far as residue N92 is concerned, its side chain is H-bonded to Tyr-c128 of the protease domain (Hopfner et al. 1999). It is possible that the N92 side chain in FIXa changes its partner upon binding to FVIIIa and N92A is unable to carry out this function. Experimentally determined FIXa/FVIIIa interface structural data are needed to assess the roles of N89 and N92 residues of FIXa. Moreover, mutations in the EGF2 domain of FIXa may affect the conformation and structure of the protease domain that has been shown to bind to FVIIIa. Thus, we emphasize that unequivocal data are needed to determine the role of the EGF2 domain in direct binding to FVIIIa.

EGF1 Domain

Several investigators have addressed the importance of the Ca^{2+} -binding site in the EGF1 domain in its interaction with FVIIIa. Studies of Rees et al. (1988) revealed that the EGF1 domain is imperative to FIXa function and that Ca^{2+} binding to this domain stabilizes the conformation that is necessary for interaction of FIXa with FVIIIa in the presence of PL. Similarly, studies of Lenting et al. (1996) showed that Ca^{2+} binding to the EGF1 domain of FIXa is essential for assembly of the FIXa/FVIIIa complex on the PL surface. However, mutation of Gln-50—the side chain of which is involved in binding to Ca^{2+} —to Pro in FIXa results in a reduced affinity for FVIIIa in the presence of PL but not in its absence (Mathur and Bajaj 1999). Thus, Ca^{2+} binding to the EGF1 domain of FIXa provides the structural elements that serve to correctly position the protease and EGF2 domains above the PL surface for optimal interaction with FVIIIa.

To assess the role of the EGF1 domain of FIXa in its direct binding to FVIIIa, investigators replaced this domain with that of protein C (Zhong et al. 1994), FVII (Chang et al. 1997), or FX (Lin et al. 1990). FIXa with the EGF1 domain from protein C ($\text{FIX}_{\text{PCEGF1}}$) had an approximate 80-fold reduction in affinity for binding to FVIIIa in a system containing PL. However, in a system without PL, it had affinity for FVIIIa equal to that of normal FIXa (Mathur and Bajaj

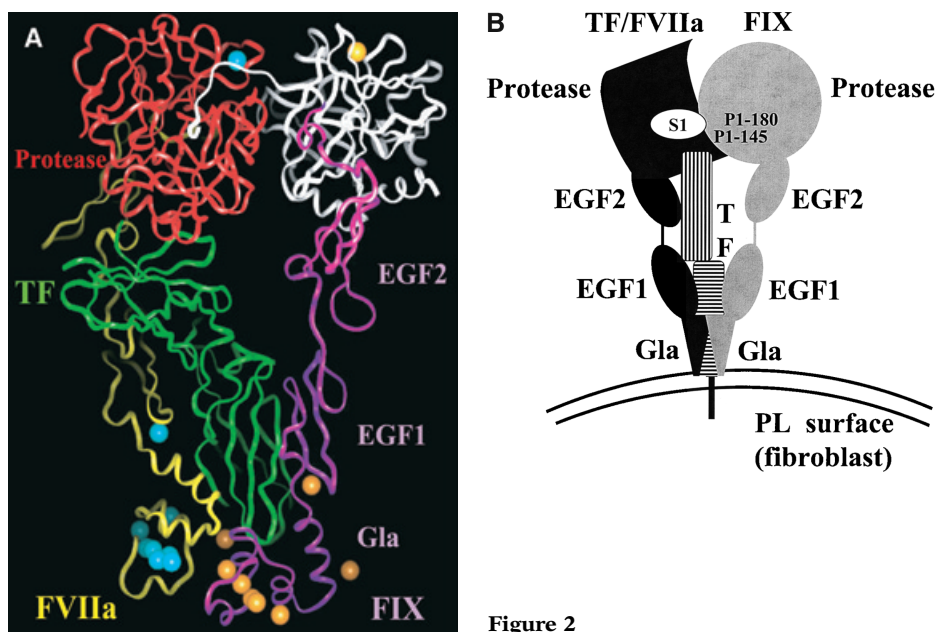


Figure 2

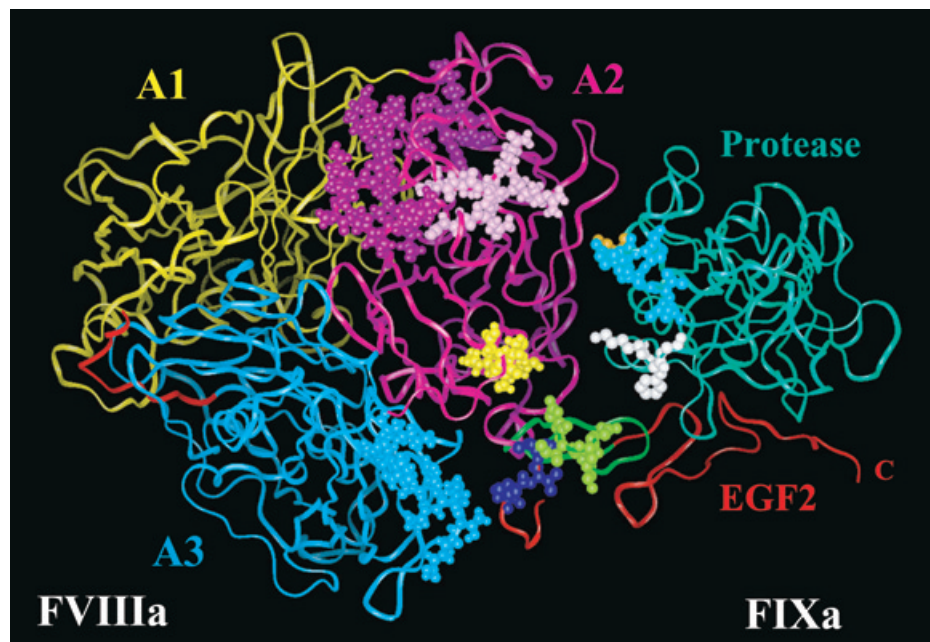


Figure 3

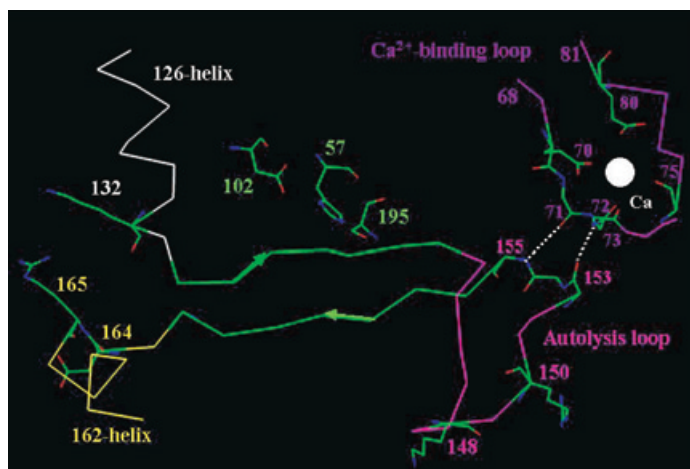


Figure 4

Figure 2. Illustration of the ternary complex of TF/FVIIa with FIX. **(A)** Ribbon representation of the FIX/TF/FVIIa ternary complex as modeled by Chen et al. (2002). TF (without the cytoplasmic tail) is in *green*, the heavy chain (protease domain) of FVIIa is in *red*, and the light chain (Gla, EGF1, EGF2) is in *yellow*. The heavy chain (protease domain) of FIX is in *white* and the light chain (Gla, EGF1, EGF2) is in *purple*. The Ca^{2+} ions bound to FVIIa are shown in *cyan* and those bound to FIX are shown in *orange*. According to this model, both the Gla and EGF1 domains of FIX bind to TF. The activation peptide segment of FIX shown is that in which the Arg¹⁴⁵-Ala¹⁴⁶ bond has been cleaved and the Arg¹⁸⁰-Val¹⁸¹ bond is positioned and poised for proteolysis in the active site of FVIIa. **(B)** Sketch representation of the TF/FVIIa complex with FIX. The sketch of the TF/FVIIa complex is based on the crystal structure determined by Banner et al. (1996), and that of zymogen FIX (Perera et al. 2001) is based on the crystal structure of porcine FIXa determined by Brandstetter et al. (1995). The association of each domain of FVIIa with the N-terminal and the C-terminal domains of TF are from Banner et al. (1996). The association of the Gla domain of FIX with TF is based on the work of Kirchhofer et al. (2000), and the association of the EGF1 domain of FIX is from Zhong et al. (1994 and 2002). The TF/FVIIa complex initially cleaves the Arg¹⁴⁵-Ala¹⁴⁶ peptide bond followed by cleavage of the Arg¹⁸⁰-Val¹⁸¹ peptide bond (Bajaj et al. 1983, Lawson and Mann 1991). Proteolysis at these two P1 positions either results from movement of the protease S1 site or the zymogen FIX cleavage site. Alternatively, TF/FVIIa may exist as a dimer (Doñate et al. 2000) with each monomeric complex catalyzing individual steps in FIX activation. The N-terminal domain of TF is shown in *white* with *vertical bars* and the C-terminal domain with a cytoplasmic tail (*vertical black line*) is shown with *black horizontal bars*. Each domain of FVIIa is shown in *black* and that of FIX is shown in *gray*. The entire assembly is shown on the fibroblast PL surface, a cell that constitutively expresses TF in the extravascular space. The P1-Arg¹⁴⁵ and P1-Arg¹⁸⁰ sites on FIX as well as the S1 site on FVIIa are indicated. F, factor; EGF, epidermal growth factor; PL, platelet phospholipid membrane; TF, tissue factor.

Figure 3. A model of the interaction of FIXa with FVIIIa. The coordinates for the human FIXa structure are from the Protein Data Bank (code 1RFN) and the coordinates for the A1, A2, and A3 subunits of FVIIIa are based on the work of Stoilova-McPhie et al. (2002). Ribbon structure for each protein is depicted. The C1 and C2 subunits of FVIIIa and the Gla and EGF1 domains of FIXa are not shown. The A1, A2, and A3 domains of FVIIIa are homologous to the ceruloplasmin domains, whereas the C1 and C2 domains are homologous to the galactose oxidase lipid-binding domain. The hypothesized interface residues are shown as CPK (Corey-Pauling-Kendrew) space-filling models. The residue Met-86 (modified from Val for expression purposes) in the human FIXa crystal structure was changed to Val using the program "O" (Jones et al. 1991). With this program, residues 78–85 (EGKNCELD) were also added to the human FIXa structure based on the coordinates of porcine FIXa (PDB code 1PFX). The FIXa protease domain is in *green* and the EGF2 domain is in *red*, with the 88–109 loop 1 and loop 2 region in *lime green*. The FIXa protease domain residue c178 (346) is in *orange*, residues of the c162-helix (330-helix) are in *cyan*, and residues of the c126-helix (293-helix) are in *white*. These protease domain residues most likely interact with the A2 subunit of FVIIIa. The FIXa EGF2 domain residues 89–93 are *lime green* and residues 84–87 are in *dark blue*. These EGF2 residues may be involved in binding to the A2 and/or the A3 subunit in FVIIIa. The A1 subunit of FVIIIa is in *yellow*, the A2 subunit is in *magenta*, and the A3 subunit is in *blue*, with the C-terminal portion that connects to the C1 subunit in *red*. The A3 subunit residues 1811–1818 are shown in *blue*. The A2 subunit residues 558–565 are in *light purple*, 511–530 are in *dark purple*, and 707–712 are in *yellow*. The A2 subunit residues could possibly be involved in binding to the heavy chain (protease domain), whereas the A3 subunit residues could be involved in binding to the light chain (including the EGF2 domain) of FIXa. F, factor; EGF, epidermal growth factor.

Figure 4. Spatial relationship of the protease domain Ca^{2+} loop (c68–c81), autolysis loop (c141–c155), and the two helices (c126-helix and c162-helix) implicated in binding FVIIIa. The Ca^{2+} loop is in *purple*, the autolysis loop is in *magenta*, the c126-helix is in *white*, and the c162-helix is in *yellow*. Residues in the figure are numbered based on chymotrypsin equivalents. The positions of the active site residues c57, c102, and c195 are indicated. The autolysis loop is connected to the c126-helix as well as to the c162-helix through a two-stranded short β sheet shown in *green*. Further, the autolysis loop is connected to the Ca^{2+} loop via two H bonds, one involving the carbonyl group of c71 and the amide group of c155, and the other involving the amide group of c73 and the carbonyl group of c153. The H bonds are shown as *white dotted lines*. The main chain carbonyl groups of c72 and c75 and the side chain carboxylates of c70 and c80 are thought to provide ligands to coordinate to Ca^{2+} . Abolishing the Ca^{2+} site by mutagenesis of Glu-c80 to Val could slightly alter the conformation of the Ca^{2+} loop, which may lengthen and weaken/disrupt the H bonds between the autolysis loop and the Ca^{2+} loop. This could lead to increased susceptibility of the autolysis loop to proteolysis. In this regard, it has been observed that specific proteolysis at c148/c149 and at c150/c151 results in diminished binding of FIXa to FVIIIa. Side chains of Lys-c148 and Arg-c150 are shown. The side chains of Lys-c132, Asp-c164, and Arg-c165 implicated in binding to FVIIIa are also depicted. Ca^{2+} cation is labeled as Ca.

1999). In biologic assays, FIXa_{VIIIEGF1} had an approximate 4-fold increase in activity (Chang et al. 1997), and FIXa_{XEGF1} had activity equivalent to normal FIXa (Lin et al. 1990). Further, the increased activity of FIXa_{VIIIEGF1} was attributed to its increased affinity for FVIIIa (Chang et al. 1997), which could be due to positioning of the protease and EGF2 domains in a more favorable orientation to interact with FVIIIa. Moreover, in a recent report, binding of a monoclonal antibody to the EGF1 domain of FIXa did not affect FVIIIa binding (Persson et al. 2002). From these studies, it is possible to conclude that the EGF1 domain of FIXa does not directly interact with FVIIIa; instead, it serves a structural role in properly orienting the EGF2 and protease domains to bind to FVIIIa.

Gla Domain

The Gla domain of FIXa has also been implicated in binding to FVIIIa. Preliminary studies reported with the Gla domain have been performed using a synthetic Gla domain (Rajotte et al. 2001). Further, the mutation of Gly-12 to Arg in the Gla domain of FIX that has been reported to impair FVIIIa binding may result from a structural impairment rather than from a direct binding effect (Larson et al. 1996). Thus, there is insufficient data to properly evaluate the role of the Gla domain in binding to FVIIIa.

• Conclusions

From the available data and literature, it appears that the Gla and EGF1 domains, as well as the activation peptide segment of FIX, contain the primary recognition determinations for binding to TF in the TF/FVIIa complex. In contrast, the protease and EGF2 domains of FIXa contain the primary recognition sites for binding to FVIIIa. Crystallographic studies of the structure of the ternary complex of FIX/TF/FVIIa and FIXa/FVIIIa are needed to precisely define the interactions in these macromolecular complexes. This represents a formidable task. However, such information is eagerly awaited, and will undoubtedly aid in development of a new class of antithrombotics, which could disrupt the interaction of FIX with TF as well as the interaction of FIXa with FVIIIa.

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